MIDSEMESTER EXAMS

MARINE ENGINEERING SYSTEMS 1

BEE27 BME 27 BNS 27

TIME ALLOWED 1.5 hours

Q1a: Briefly describe how the torque is transmitted to the rudder stock in a rotary vane steering gear.

B: Itemise the tests you would carry out on a steering gear before leaving port.

Q2a: With reference to hydraulic steering gears explain why relief valves are provided as well as shock valves,

b; Explain the purpose of the hunting gear.

Q3a: Water for boilers is usually kept as pure as reasonably possible, give reasons why this is so.

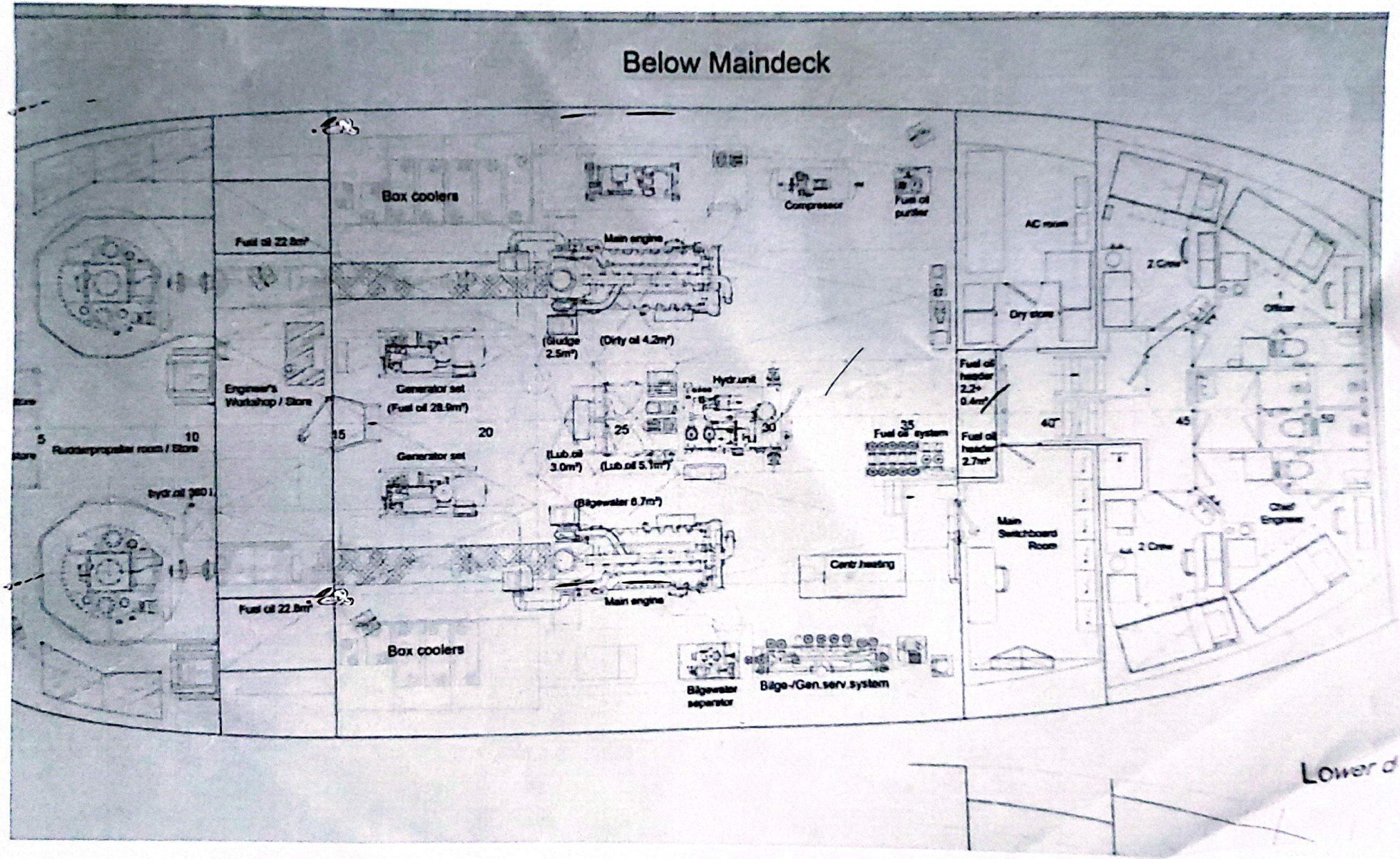
b; Briefly describe why boiler water needs to be tested periodically and state two of the tests.

C; Why is it necessary to keep oxygen out of the boiler, how is this done mechanically and chemically.

Q4: Explain in detail the Six(6) items on the General Arrangement drawing of the boat attached.

rudder

CamScanner


Below Maindeck
Box coolers
Compressor
Fuel oil purifier
AC room
Fuel oil 22.8m³
Main engine
2 Crew
Dry store
Officer
(Sludge 2.5m³)
(Dirty oil 4.2m³)
Engineer's Workshop / Store
Generator set
(Fuel oil 28.9m³)
Hydr.unit
Fuel oil header 2.2+ 0.4m³
Fuel oil header 2.7m³
Fuel oil system
5
10
15
20
25
30
35
40
45
Rudderpropeller room / Store
Generator set
(Lub.oil 3.0m³)
(Lub.oil 5.1m³)
hydr.oil 380 l
(Bilgewater 6.7m³)
Main Switchboard Room
Chief Engineer
2 Crew
Centr.heating
Fuel oil 22.8m³
Main engine
Box coolers
Bilgewater separator
Bilge-/Gen.serv.system
Lower d